

Get Your Directory for
Rs ~~1000~~ Rs 750
Save 25%*

Most Trusted[#] Electronics Industry Directory Up For Grabs!!

- 10,000+ Electronics Organisations
- 2,800+ Product Categories For Buyers
- Key Government Organisations Related to Electronics
- Key Industry Associations Related to Electronics

* Postage FREE till 15th September 2018

Verified Data



Fill in the photocopy of this form and send it to:

EFY Enterprises Pvt Ltd, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110 020
Ph: 011-40596600 or E-mail: info@efy.in

ORDER FORM

Name _____

Mailing Address _____

City _____ Pin Code _____

State _____ Phone No. _____

Email _____

Please find enclosed a sum of Rs _____ by DD/MO/crossed cheque
bearing the No. _____ drawn on (bank) _____

in favour of EFY Enterprises Pvt Ltd, payable at Delhi (Please add Rs 50 for
non-metro cheques) OR Pay online at: <http://payment.efyindia.com>

and microphone, for example. This threat perception (or lack of thereof) fits squarely into the colloquial notion of what eavesdropping actually entails. It opens the doors for a plethora of rather sinister attacks that can be carried out with the user actually allowing these by granting sensor data access permissions.

Let us survey some of the examples that are well known in the academic community.

Device finger-printing. Most commercial smartphone sensor chips provide measurements that slightly deviate from the ground truth values in a way that is rather specific to that particular device. This is due to the idiosyncratic birth defects or hardware imperfections etched into the silicon. These device-specific defects manifest as offsets and scaling factors in some measurements, or in terms of the measurements that these yield in response to some pre-authored triggers controlled by the attacker.

By using the right algorithm, the attacker can very accurately finger-print the device, allowing it to be tracked even when the user has taken standard precautions such as using incognito modes in the browser or deleting suspicious cookies. The sinister aspect of this attack is that, it empowers the attacker to keep consolidating the data collected from the victim and track her or him over disparate spatial locations and time. The attack becomes ineffective if and only if the device is changed or the sensor ages or wears out dramatically to the degree that its fingerprint becomes obsolete. But studies have shown that the fingerprint lasted over nine months of experimentation with more than 100 accelerometers.

Acoustic (voice) data collection without using the microphone. Two factors facilitate this effect. The first factor is that both the accelerometer and gyroscope are extremely sensitive to any mechanical vibration including human speech. Second, both iOS and Android allow high enough sampling rates of data from these sensors that a non-trivial part of the human speech spectrum is covered.

Projects such as Gyrophone and Accelword demonstrate how gyroscopic and accelerometric vibration data can be used to achieve efficient speaker information retrieval or hot-word detection. What makes it truly worse is that sensor data can be extracted via browsers as well, albeit at a lower sampling rate.

In a recent project titled 'PINlogger,' researchers demonstrated that once the victim visits a specific malicious website owned by the attacker, a simple JavaScript code embedded in the Web page can be used to trigger motion and orientation sensor data collection that does not need any permission from the user. These sensor streams are then analysed to infer the victim's 4-digit PINs with 94 per cent accuracy in just three attempts!

On the concluding note, I would like to emphasise that this article is not meant to introduce fearmongering in the minds of smartphone users with regards to IMU motion sensors, but to merely raise awareness about the sinister threats posed by the misuse of these sensors. These sensors will be at the heart of some exciting advancements spanning across domains such as healthcare, authentication, entertainment and videography. **EFY**